CHECK WHAT YOU LEARNT

Check Yourself - Unit 5

A. Choose the correct answer

1. The left-hand rule fails on a maze that is
   (a) large (b) non-simply-connected (c) dark (d) square
2. In a state machine the system is always in
   (a) several states (b) exactly one state (c) no state (d) a loop
3. The path LBR simplifies to
   (a) L (b) B (c) R (d) S
4. simplify() is called after every turn in order to
   (a) speed up the motors (b) keep the stored path short (c) improve the sensors (d) reduce current
5. The second run is faster because the robot
   (a) drives faster (b) no longer decides at junctions (c) has fewer sensors (d) uses less power
6. A layer that refuses an unsafe action regardless of the main logic is a
   (a) state machine (b) safety interlock (c) median filter (d) bootloader

B. Fill in the blanks

1 Each turn is stored as a single ______________: L, R, S or B.

2 The four states in this design are FOLLOWING, AT_JUNCTION, ______________ and FINISHED.

3 Proving each subsystem alone before combining them is called ______________ testing.

4 IR sensors should be mounted ______ to ______ mm above the floor.

C. Answer in one or two lines

1 Explain the left-hand rule and state honestly when it fails.

2 Write out a route your robot walked and show one simplification step.

3 Explain why a state machine is better than a chain of if statements here.

4 Give your first-run and second-run times, and account for the difference.

Extra Challenges

For groups whose builds are working early. All use parts you already have.

#	Challenge	Uses
1	Modulate the laser and accept only light pulsing at that rate	laser + LDR + timing
2	Timestamp every intrusion and show the last three on the LCD	millis() + array
3	An armed and disarmed mode, toggled by the push switch	state machine
4	Log 50 readings at each of five distances and report the standard deviation	HC-SR04 + maths
5	Compensate for air temperature using the DHT11 formula	speed of sound
6	Plot live distance on the Serial Plotter, raw against filtered	Serial Plotter
7	A servo-mounted sensor that sweeps and reports the clearest direction	HC-SR04
8	Print the simplified path to the LCD at the end of a run	LCD + array
9	A safety interlock that stops the robot before any wall contact	HC-SR04
10	Compare left-hand and right-hand rules on the same maze and report which won	test data

MIXED REVISION

Twenty Questions to Test Yourself

1. Write a testable requirement for an alarm's response time.
2. Distinguish accuracy from precision with an example of each.
3. Which kind of error can be corrected in software, and which must be filtered?
4. How would you estimate the uncertainty of a measurement from data alone?
5. Which chip on the Arduino Uno is the actual computer, and what clocks it?
6. How much flash and how much SRAM does it have, and what lives in each?
7. Describe the four stages between pressing Upload and your program running.
8. How many distinct values can a 10-bit ADC report, and what is one step worth?
9. Explain successive approximation in one sentence.
10. Why is delay() unacceptable in a security system?
11. How does a laser diode differ from an LED in the light it produces?
12. Explain photoconductivity in an LDR.
13. Why does latching matter for a fast beam-break event?
14. How does a piezoelectric disc work as both speaker and microphone?
15. State the beam angle, blind zone and maximum range of an HC-SR04.
16. How is a gain correction derived from two data points?
17. Why does a median filter beat an averaging filter for ultrasonic data?
18. State the left-hand rule and one maze that defeats it.
19. Explain the simplification LBS becomes R.
20. Name the design decision in your capstone you are least confident about, and why.

Wiring Reference - All Three Projects

Keep this page open while you build. Every pin used this year, in one place.

Project 1 - Laser Security System

Arduino pin	Connects to
A0	LDR module AO
D8	Buzzer +
D7	Push switch, with 10K to GND
D12 / D11	LCD RS / E
D5, D4, D3, D2	LCD D4, D5, D6, D7
-	Laser + to 5V through 220 ohm, - to GND
-	LCD V0 to potentiometer wiper; VSS, RW, K to GND; VDD to 5V

Project 2 - Ultrasonic Measurement and Calibration

Arduino pin	Connects to
D9	HC-SR04 TRIG
D10	HC-SR04 ECHO
D12 / D11	LCD RS / E
D5, D4, D3, D2	LCD D4, D5, D6, D7
5V and GND	HC-SR04 VCC and GND
-	LCD V0 to potentiometer wiper

Project 3 - Maze Solver Robot

Arduino pin	Connects to
D2 / D3 / D4	Left / centre / right IR sensor OUT
D11 / D12	HC-SR04 TRIG / ECHO
D8, D7	L298N IN1, IN2 - left motor direction
D6, D5	L298N IN3, IN4 - right motor direction
D10, D9	L298N ENA, ENB - speed
GND	L298N GND - the shared ground, never omit
-	L298N 12V and GND to the battery pack
-	L298N OUT1-4 to the two motors

Words to Know

Word	Meaning
Accuracy	How close a reading is to the true value.
ADC	The circuit that converts a voltage into a number.
ATmega328P	The microcontroller that is the real computer on the board.
Beam angle	The cone within which an ultrasonic sensor can see.
Blind zone	The close range where a sensor cannot measure at all.
Bootloader	A resident program that writes your code into flash.
Collimated	Formed into a narrow parallel beam.
Debounce	Ignoring the brief contact chatter of a switch.
False positive	An alarm with nothing to alarm about.
False negative	No alarm when there really was something.
Gain error	Error that grows in proportion to the reading.
H-bridge	Four switches that let a circuit reverse a motor.
Independent testing	Proving each subsystem alone before integrating.
Latching	Staying on once triggered, until deliberately cleared.
Left-hand rule	Always prefer the leftmost available turn.
Median filter	Taking the middle of several sorted readings.
Modulation	Pulsing a signal at a known rate so it cannot be imitated.
Non-blocking	Code that waits without halting everything else.
Offset error	A constant shift present in every reading.
Photoconductivity	Light freeing electrons and lowering resistance.
Piezoelectric	Changing shape under voltage, and producing voltage when squeezed.
Precision	How closely repeated readings agree with each other.
Random error	Unpredictable scatter, reducible by averaging.
Requirement specification	A written, testable statement of what a system must do.
Resolution	The smallest change a measuring system can detect.
Safety interlock	A layer that refuses an unsafe action regardless of the main logic.
Shared ground	The wire letting two supplies agree on what zero volts means.
State machine	A design where the system is always in exactly one named state.
Successive approximation	The binary-search method an ADC uses.
Systematic error	A consistent offset or drift that can be corrected.
Uncertainty	The range within which the true value very likely lies.

Answer Key

Section A only. Sections B and C are for you and your trainer to discuss - there is often more than one defensible answer.

Check Yourself - Unit 1

Q	Answer	The correct option
1	(b)	testable
2	(b)	systematic error
3	(b)	precision
4	(b)	averaging several readings
5	(b)	40 and 44
6	(b)	13.6 mA

Check Yourself - Unit 2

Q	Answer	The correct option
1	(b)	the ATmega328P
2	(b)	SRAM
3	(c)	1024 values
4	(b)	4.9 millivolts
5	(b)	successive approximation
6	(c)	unsigned long

Check Yourself - Unit 3

Q	Answer	The correct option
1	(b)	coherent and collimated
2	(b)	very high
3	(b)	slow to settle
4	(b)	a fast event could be missed between loops
5	(b)	false negative
6	(b)	modulated

Check Yourself - Unit 4

Q	Answer	The correct option
1	(c)	both, in opposite directions
2	(b)	just above human hearing
3	(b)	blind zone
4	(b)	gain error
5	(b)	the mean more than the median
6	(b)	warmer

Check Yourself - Unit 5

Q	Answer	The correct option
1	(b)	non-simply-connected
2	(b)	exactly one state
3	(b)	B
4	(b)	keep the stored path short
5	(b)	no longer decides at junctions
6	(b)	safety interlock

My Engineering Log

Fill one row after every class, including the sessions where nothing worked. A log of failures and their diagnoses is assessed alongside your hardware.

Session	What I built or measured	What failed	Diagnosis and fix
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

A last word

Look back at what changed this year. You stopped asking whether a machine works and started asking how well, within what range, and with what uncertainty. You wrote requirements before you wired anything, and you recorded failures instead of hiding them. Those habits, far more than any single project, are what engineering actually is.